



Advanced AI-Driven Search

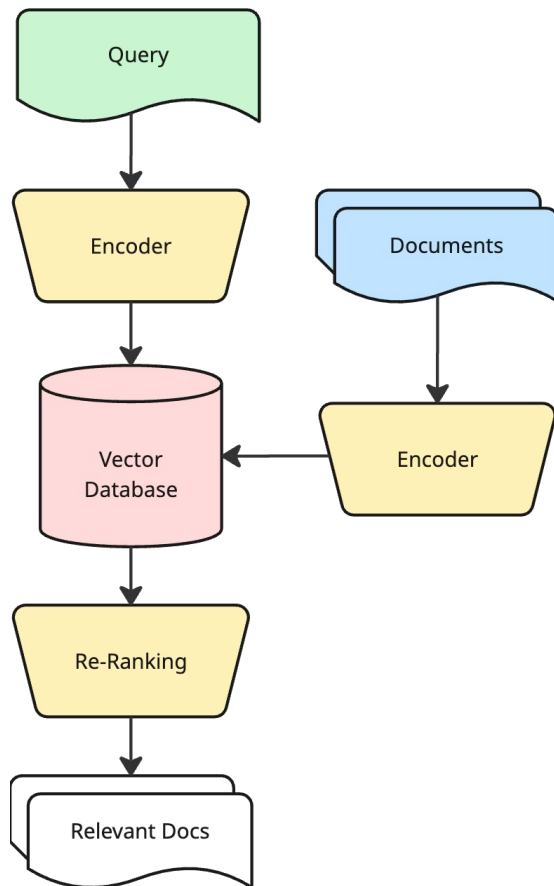
Jakub Žovák

Motivation

Information Retrieval in AI Stack

- Search based on document similarity
- Use vector representation of documents
 - Expresses semantic meaning
 - Encoder models like BERT
- Vector databases as main engine
 - **Qdrant**, Milvus ...
- Optional re-ranking
- Business Case
 - **Context Retrieval, RAG, Image Search, Recommender**
 - This pipeline is usually not enough :(

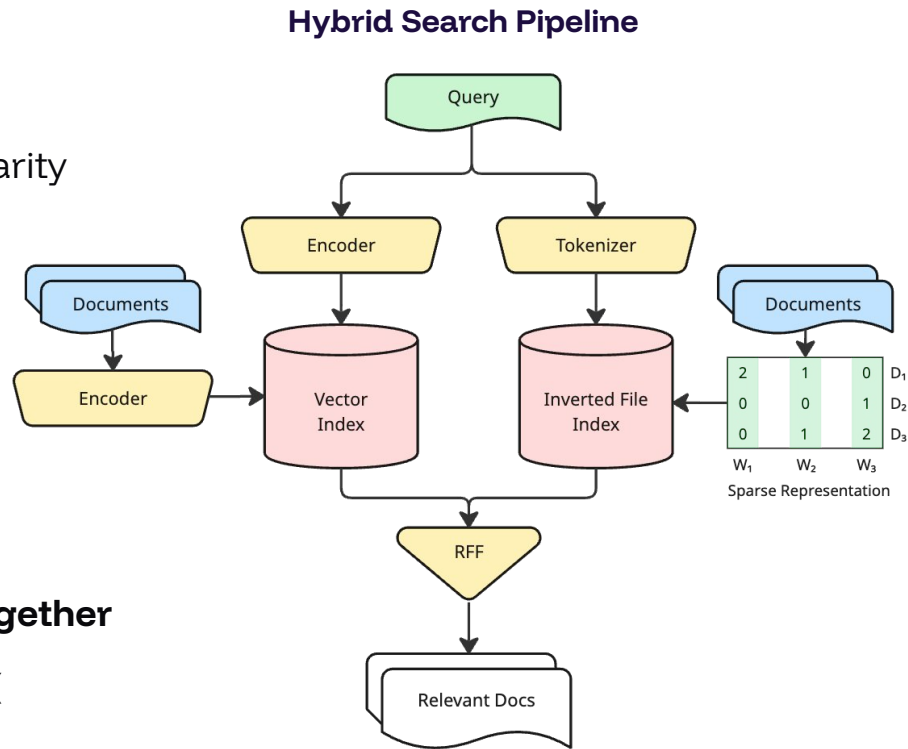
Simple Search Pipeline



Motivation

Hybrid Search - Current Standard

- Vector search
 - Embeddings + HNSW index + cosine similarity
- Keyword search
 - TF-IDF + Inverted File index + BM25
- Combination of two
 - Reciprocal Rank Fusion (RFF)
- Business Case
 - **Combines keyword and vector search together**
 - **Robust search, but there are still issues :(**



Motivation

Deficiencies of Standard Methods

Vector Search

- **Low Interpretability**
- Out of domain performance
- No exact match
- **No metadata information**

Keyword Search

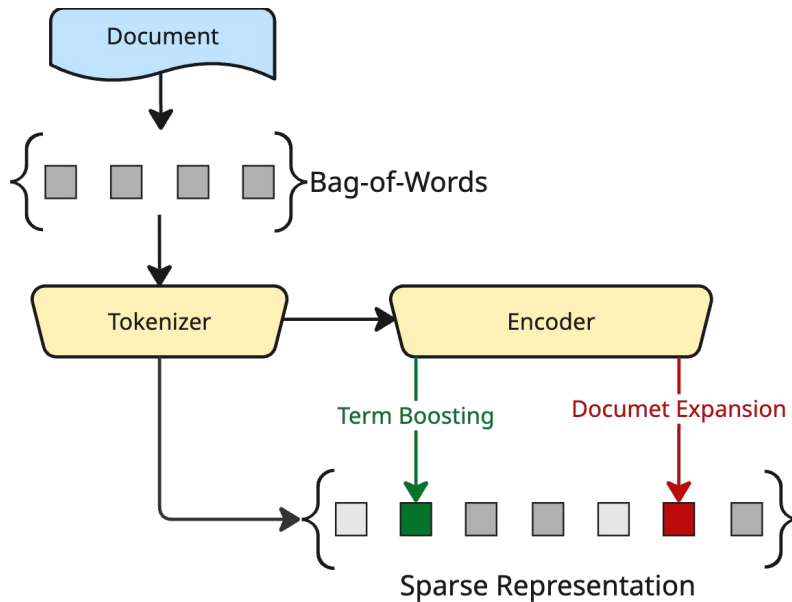
- No fine tuning
- **Vocabulary mismatch**
 - Synonyms
 - Homonyms
- No semantic understanding

Learned Sparse Retrieval (Keyword Search)

Introduction

- Business Request
 - **Synonyms and homonyms support**
 - **Keywords should represent semantics**
- Enriching sparse vectors with contextual info.
- Retains keyword search capabilities
- Two main features
 - Contextual term boosting
 - Document expansion
- Examples of learned sparse models
 - TILDE, SPARTA, **SPLADE**

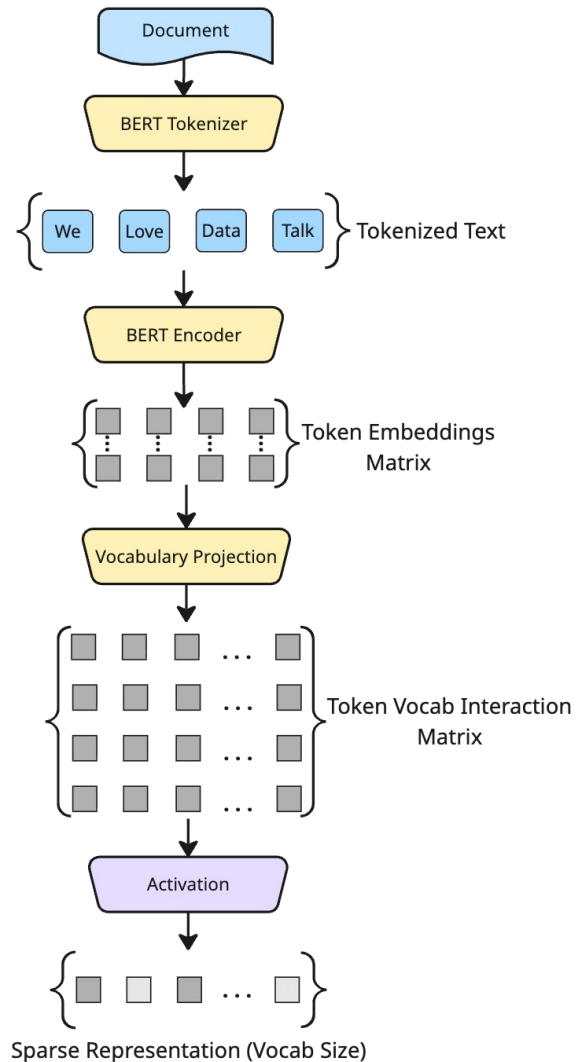
Idea Behind Learned Sparse Retrieval



Learned Sparse Retrieval (Keyword Search)

SPLADE

- Leverages BERT model
 - Adds trainable network on top
- Sparse vector dimension = vocabulary size (30522)
- Document expansion out of the box
- Trade off
 - Increased query and indexing time
- Business Case
 - **Supports synonyms & homonyms**
 - **Keywords carry semantic meaning**

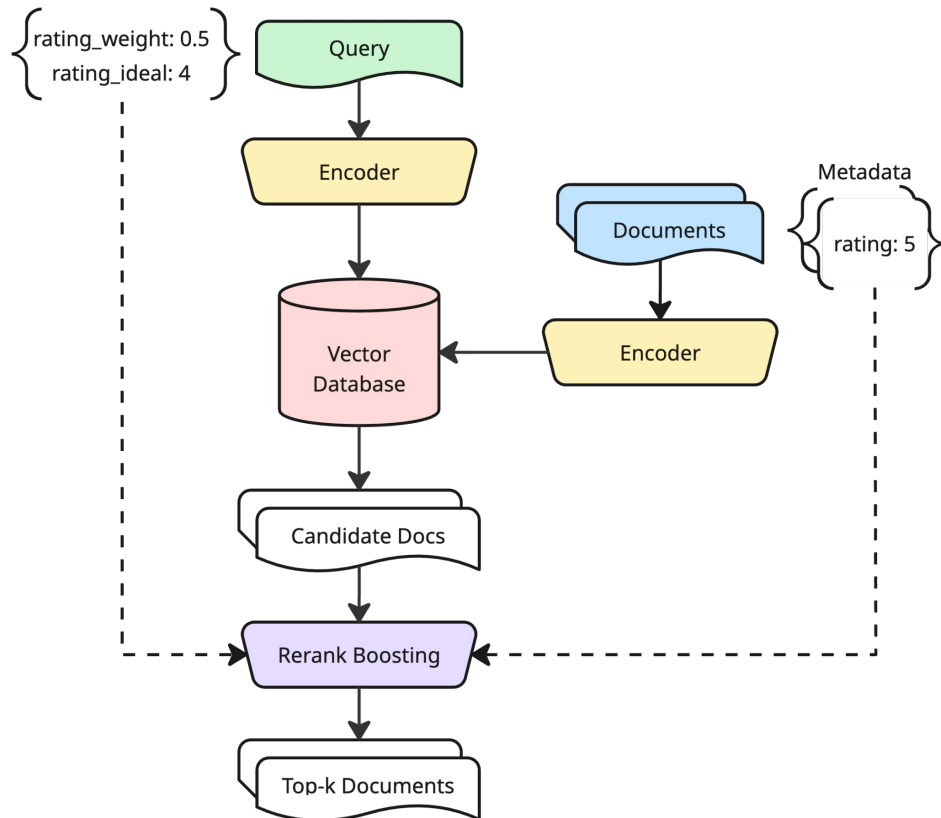


Advanced Vector Search

Metadata Boosting

- Business Request
 - **Prioritize documents by metadata**
 - E.g. Search for most similar book with highest rating
- Search for custom measure like
 - **$\text{cosine}(\text{query}, \text{doc}) + \text{rating} \times 0.5$**
- Naive re-ranking approach
 - Perform vector search
 - Rerank candidate set
- Does not scale and misses lot of docs :(

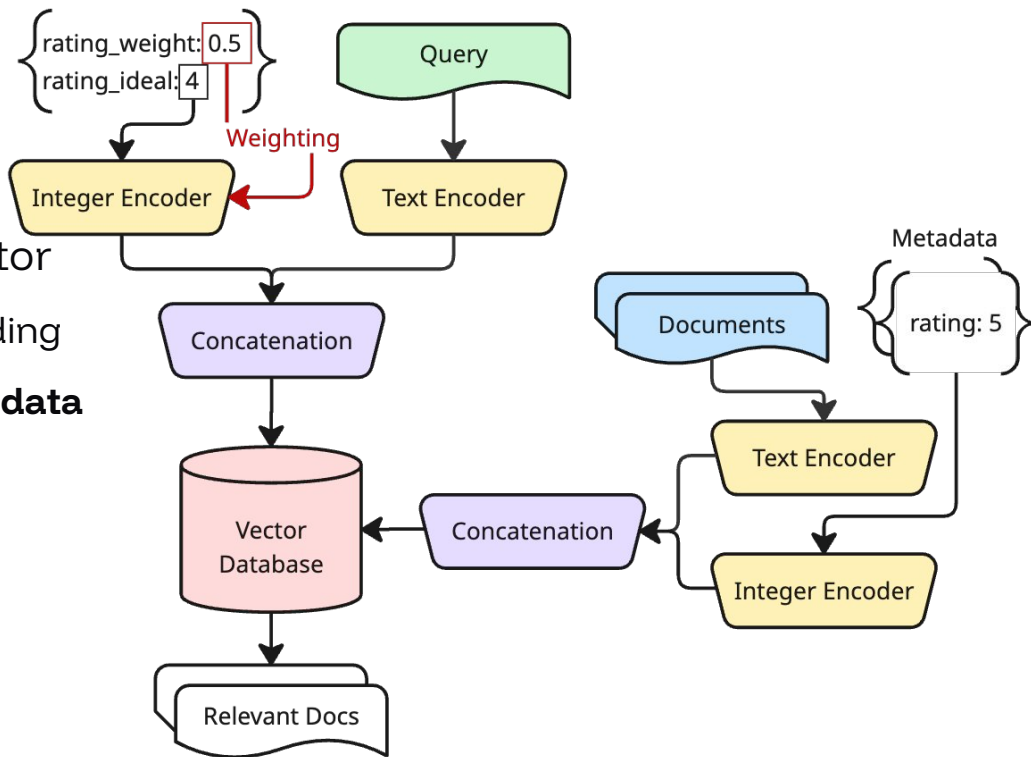
Metadata Prioritization Via Reranking



Advanced Vector Search

Mixture Of Encoders

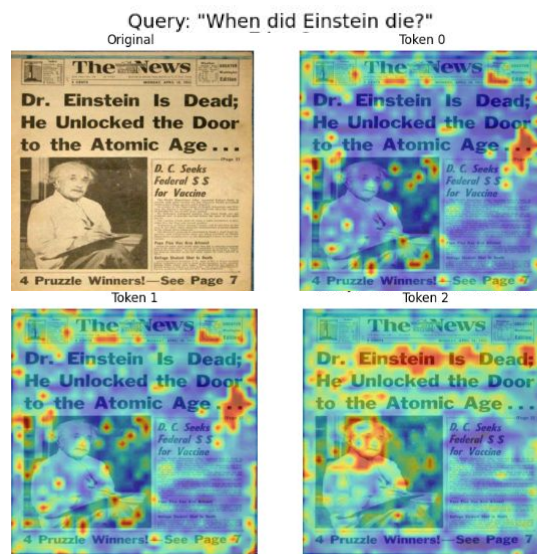
- Implemented by [SuperLinked](#)
- Encode custom measure into a vector
 - $\text{cosine}(q, d) + \text{rating} \times 0.5 \rightarrow \text{embedding}$
 - **Embeddings enhanced with metadata**
- Specific encoder per data type
- Use just vector search
 - Scalable, finds all relevant objects
- Business Case
 - **Metadata prioritization during search**
 - Support for more complex queries



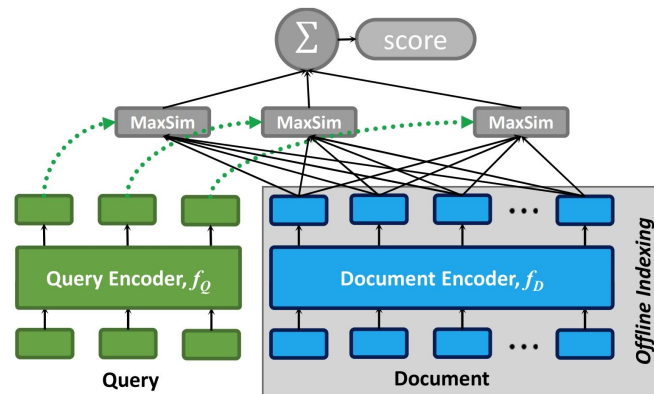
Advanced Vector Search

Multi-vector Search

- One vector per document
 - Information loss
 - No interpretability
- Multi-vector representation
 - Use of token embeddings or multiple vectors
 - Employs **MaxSim operator**
- Increased storage and computation
- Business Case
 - **Increased interpretability & expressive power**
 - Fast precomputed reranking



Visual Interpretability of ColPali

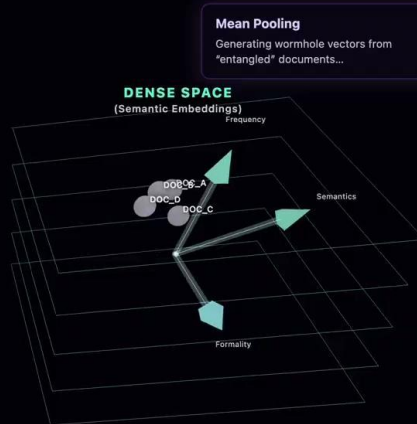
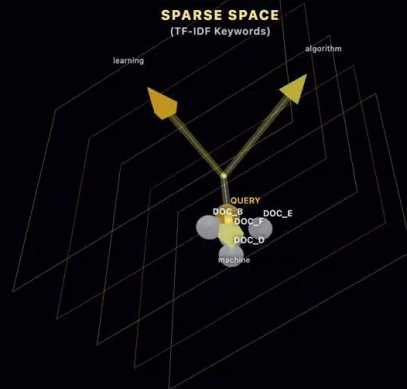


Wormhole Vectors

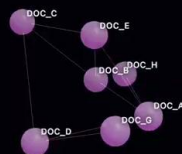
Bonus Approach

Wormhole Vectors

Query in **sparse space** → match documents → generate **entangled vector** in **dense space** → find entagled entry point to **graph representation**.



GRAPH SPACE (Knowledge Graph)



Thank you for your attention!

- Any questions?
- We are hiring
 - [AI Architect](#)
 - Skilled in AI/ML
 - Also proficient in SWE
 - AI Engineers
- Feel free to connect with me on [LinkedIn](#):

